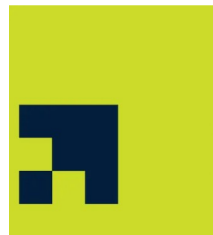


M504 AI and Applications

Assessment Brief Overview

Dr. Sanjit Kumar Saha



Gisma
University
of Applied
Sciences

Assessment Overview

- 70% Individual Project
- 15% Online Assessments
- 15% Class Participation
- Deadline: September 18, 2026 at 18:00 Berlin Time

Your Role

- Act as a Data Science Consultant
- Analyze a real-world dataset
- Answer business questions
- Present actionable insights

Submission Requirements

- Submit a Jupyter Notebook as HTML
- Upload on Canvas
- Notebook size below 20,000 characters
- Accept Declaration of Authorship

Project Structure

- Business Context
- Data Exploration
- Data Preprocessing
- Explanatory Data Analysis
- Final Discussion and Conclusion

Business Context

- Who is the client company?
- What type of data is available?
- Why is analysis important?
- Define the business problem clearly

Data Exploration

- Understand dataset characteristics
- Check missing values
- Identify duplicates and anomalies
- Explore patterns and metadata

Data Preprocessing

- Handle missing values
- Convert data formats
- Create new features
- Prepare data for analysis

Explanatory Data Analysis

- Define 5–10 meaningful business questions
- Use Python and Pandas
- Explain business importance
- Interpret results clearly

Example Business Questions

- Best month for sales?
- Top-performing products?
- Best time for advertising?
- Frequently bought products together?

For Each Question

- Add subsection in notebook
- Explain importance
- Write runnable Python code
- Interpret findings and recommendations

Final Discussion & Conclusion

- Summarize key insights
- Discuss strengths and limitations
- Provide business recommendations
- Connect analysis to decision-making

Academic Integrity

- Work must be your own
- Avoid plagiarism
- Understand every line of code
- Follow Harvard referencing

Marking Criteria

- 35% Code correctness and completeness
- 35% Report structure and writing quality
- Critical evaluation of results
- Professional presentation

Tips for High Marks

- Choose meaningful datasets
- Use good visualizations
- Write clear explanations
- Think like a consultant

Closing

- Start early and plan carefully
- Ask questions whenever needed
- Focus on both analysis and communication
- Good luck with your project!